

# Oscar Antonio Juarez Jimenez

## Artificial Intelligence Engineer

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I am an Artificial Intelligence Engineer with experience as a Data Engineer, specializing in Large Language Models, Retrieval-Augmented Generation (RAG), and AI-powered software solutions. Experienced in designing AI systems that improve information access, automate workflows, and enhance decision-making. Passionate about applying responsible AI to solve real-world challenges in public service, education, and mission-driven organizations, solving complex problems, mastering emerging technologies, and contributing to high-impact projects by bridging logic, analytics, and creativity.

## Education

### National Polytechnic Institute (IPN)- Escuela Superior de Computo

México, CDMX. Aug 2021- Jul 2025

Bachelor's Degree in Artificial Intelligence Engineering

GPA: 9.5/10

Relevant Courses: Machine Learning, Digital Image Processing, Natural Language Technologies, Neural Networks and Deep Learning, Advanced Neural Networks, Software Engineering for Intelligent Systems, Parallel Computing, Bio-Inspired Algorithms.

## Professional Experience

### Data Engineer | Bluetab - BBVA

Jul 2025 – Present

- Implemented data ingestion, transformation, and storage (ETL) processes, seamlessly integrating multiple sources into the bank's core data infrastructure.
- Managed datasets and tables in AWS S3, ensuring highly structured storage and efficient access for Spark processing and advanced data analytics.
- Maintained and optimized data pipelines in distributed computing environments (clusters), guaranteeing data reliability, quality, and high availability.
- Collaborated on data integration strategies for analytical use cases and Machine Learning models within the structural risk division, fostering the development of data-driven solutions across the banking ecosystem.

### Innovation and Development Intern | Grupo Salinas

Feb – Jul 2025

- Contributed to the research, development, and deployment of artificial intelligence tools aimed at optimizing business workflows and validating models in innovation-driven environments.
- Participated in the engineering of AI agents, conversational chatbots, visual prototypes, and dynamic landing pages.

### Banco de México | Full-Stack Intern

Jun - Dec 2024

- Boosted system performance by 30% by optimizing SQL queries and reducing algorithmic complexity in Java/JavaScript, while enhancing UX through a streamlined, intuitive interface.
- Optimized SQL database management by enforcing integrity and security via normalization, stored procedures, views, indexing, and role-based access control.

## Projects

### Object Segmentation System for Indoor 3D Model Editing Using NeRF

Jun 2024 – Aug 2025

- Designed an AI-based object segmentation system for editing 3D models in interior environments.
- Boosted system classification accuracy by 30% by integrating Machine Learning algorithms into the initial U-Net architecture.
- Achieved photorealistic 3D editing by combining NeRF's spatial reconstruction with diffusion-based texture synthesis (Stable Diffusion).

### RAG ChatBot

Jan – Jun 2024

- I collaboratively developed a Retrieval-Augmented Generation (RAG) system for LLM to generate summaries and answer natural language queries about Mexico's Diario Oficial de la Federación (DOF).
- Used LangServe and LangChain for model retrieval and Pinecone for vector storage.

To check out more of my projects, please feel free to visit my own portfolio at <https://oscarjuji.github.io/>

## Courses & Certifications

- Legacy Responsive Web Design Challenges, Legacy Python for Everybody & JavaScript Algorithms and Data

Structures – FreeCodeCamp

- Learn AI Skills Challenge (Azure) – Microsoft
  - Google Cloud Computing Foundations Certificate
  - Google Cloud GenAI Skill Badge Pathways
  - Oracle Next Education - Alura LATAM
  - Secure Development 3.0 – Net4skills
  - Jira Scrum Team
  - AWS Technical Essentials
  - SAFe Scrum Master Exam by Scaled Agile
  - AWS Certified Data Engineer – Associate (\*In Progress)
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### **Skills**

**Programming Languages: Intermediate (2 years):** C/C++, Python, Java, JavaScript, SQL, and Excel; Basic ( $\leq 1$  year): C#, CUDA, Go, PHP, TypeScript, R, Scala.

**Tools & Technologies: Intermediate (2 years):** Git, Google Cloud Platform (GCP), Docker, HTML/CSS, ☑ Large Language Models, Retrieval-Augmented Generation, prompt Engineering, LangChain, LangServe, pinecone, computer vision, deep learning neural Networks, machine Learning, Tableau; Basic (1 year): Power BI, MLflow, Kubernetes, Azure, Amazon Web Services (AWS).